MEDIUM · ADVANCED MATH · ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

05

10 answers · Rationale-rich · Teach from doc

Q1 either 7

The correct answer is either 7, 8, or 13. Since the given equation has two integer solutions, the expression on the left-hand side of this equation can be factored as $(x+c)(x+d)$, where c and d are also integers. The product of c and d must equal the constant term of the original quadratic expression, which is 12. Additionally, the sum of c and d must be a negative number since it's given that $a>0$, but the sign preceding a in the given equation is negative. The possible pairs of values for c and d that satisfy both of these conditions are -4 and -3, -6 and -2, and -12 and -1. Since the value of $-a$ is the sum of c and d, the possible values of $-a$ are $-4+(-3)=-7$, $-6+(-2)=-8$, and $-12+(-1)=-13$. It follows that the possible values of a are 7, 8, and 13. Note that 7, 8, and 13 are examples of ways to enter a correct answer.

Q2 C

C. 2

Choice C is correct. The solutions to a system of two equations correspond to points where the graphs of the equations intersect. The given graphs intersect at 2 points; therefore, the system has 2 solutions.

Choice A is incorrect because the graphs intersect. Choice B is incorrect because the graphs intersect more than once. Choice D is incorrect. It's not possible for the graph of a quadratic equation and the graph of a linear equation to intersect more than twice.

Q3

A

A. Exactly two

Choice A is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a, b, and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $x^2 - 12x + 27 = 0$, $a = 1$, $b = -12$, and $c = 27$. Substituting these values for a, b, and c in $b^2 - 4ac$ yields $-12^2 - 4127$, or 36. Since the value of its discriminant is positive, the given equation has exactly two distinct real solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

B

B. 2

Choice B is correct. Since $\frac{x+5}{x+5}$ is equivalent to 1, the right-hand side of the given equation can be rewritten as $\frac{x+5}{x+5}-\frac{1}{x+5}$, or $\frac{x+4}{x+5}$. Since the left- and right-hand sides of the equation $\frac{2(x+1)}{x+5}=\frac{x+4}{x+5}$ have the same denominator, it follows that $2(x+1)=x+4$. Applying the distributive property of multiplication to the expression $2(x+1)$ yields $2(x)+2(1)$, or $2x+2$. Therefore, $2x+2=x+4$. Subtracting x and 2 from both sides of this equation yields $x=2$.

Choices A, C, and D are incorrect. If $x=0$, then $\frac{2(0+1)}{0+5}=1-\frac{1}{0+5}$. This can be rewritten as $\frac{2}{5}=\frac{4}{5}$, which is a false statement. Therefore, 0 isn't a solution to the given equation. Substituting 3 and 5 into the given equation yields similarly false statements.

Q5

B

B. $a=\frac{x}{8b+9}$

Choice B is correct. To express a in terms of b and x, the given equation can be rewritten such that a is isolated on one side of the equation. Since it's given that b is a positive number, $b+9$ is not equal to zero. Therefore, dividing both sides of the given equation by $8b+9$ yields the equivalent equation $\frac{x}{8b+9}=a$, or $a=\frac{x}{8b+9}$.

Choice A is incorrect. This equation is equivalent to $x=8a+b+9$.

Choice C is incorrect. This equation is equivalent to $x=\frac{8b+9}{a}$.

Choice D is incorrect. This equation is equivalent to $x=\frac{a}{8b+9}$.

Q6

5

The correct answer is 5. Multiplying both sides of the given equation by $x + 6$ results in $55 = xx + 6$. Applying the distributive property of multiplication to the right-hand side of this equation results in $55 = x^2 + 6x$. Subtracting 55 from both sides of this equation results in $0 = x^2 + 6x - 55$. The right-hand side of this equation can be rewritten by factoring. The two values that multiply to -55 and add to 6 are 11 and -5. It follows that the equation $0 = x^2 + 6x - 55$ can be rewritten as $0 = x + 11x - 5$. Setting each factor equal to 0 yields two equations: $x + 11 = 0$ and $x - 5 = 0$. Subtracting 11 from both sides of the equation $x + 11 = 0$ results in $x = -11$. Adding 5 to both sides of the equation $x - 5 = 0$ results in $x = 5$. Therefore, the positive solution to the given equation is 5.

Q7

A

A. $H = 1 - \frac{V_P}{V_B}$

Choice A is correct. The hematocrit can be expressed in terms of the blood volume and the plasma volume by solving the given equation $V_B = \frac{V_P}{1 - H}$ for H. Multiplying both sides of this equation by $(1 - H)$ yields $V_B(1 - H) = V_p$. Dividing both sides by V_B yields $1 - H = \frac{V_P}{V_B}$. Subtracting 1 from both sides yields $-H = -1 + \frac{V_p}{V_B}$. Dividing both sides by -1 yields $H = 1 - \frac{V_P}{V_B}$.

Choices B, C, and D are incorrect and may result from errors made when manipulating the equation.

Q8

A

A. –1

Choice A is correct. It is given that $y = x + 1$ and $y = x^2 + x$. Setting the values for y equal to each other yields $x + 1 = x^2 + x$. Subtracting x from each side of this equation yields $x^2 = 1$. Therefore, x can equal 1 or –1. Of these, only –1 is given as a choice.

Choice B is incorrect. If $x = 0$, then $x + 1 = 1$, but $x^2 + x = 0^2 + 0 = 0 \neq 1$. Choice C is incorrect. If $x = 2$, then $x + 1 = 3$, but $x^2 + x = 2^2 + 2 = 6 \neq 3$. Choice D is incorrect. If $x = 3$, then $x + 1 = 4$, but $x^2 + x = 3^2 + 3 = 12 \neq 4$.

Q9

28

The correct answer is 28. The given absolute value equation can be rewritten as two linear equations: $4x - 4 = 112$ and $-4x - 4 = 112$, or $4x - 4 = -112$. Adding 4 to both sides of the equation $4x - 4 = 112$ results in $4x = 116$. Dividing both sides of this equation by 4 results in $x = 29$. Adding 4 to both sides of the equation $4x - 4 = -112$ results in $4x = -108$. Dividing both sides of this equation by 4 results in $x = -27$. Therefore, the two values of $x - 1$ are $29 - 1$, or 28, and $-27 - 1$, or -28. Thus, the positive value of $x - 1$ is 28.

Alternate approach: The given equation can be rewritten as $4x - 1 = 112$, which is equivalent to $4x - 1 = 112$. Dividing both sides of this equation by 4 yields $x - 1 = 28$. This equation can be rewritten as two linear equations: $x - 1 = 28$ and $-x - 1 = 28$, or $x - 1 = -28$. Therefore, the positive value of $x - 1$ is 28.

Q10

99

The correct answer is 99. In the given system of equations, the second equation is $x + 8 = 11$. Subtracting 8 from both sides of this equation yields $x = 3$. In the given system of equations, the first equation is $y = x^2 + 14x + 48$. Substituting 3 for x in this equation yields $y = 3^2 + 143 + 48$, or $y = 99$. Therefore, the solution to the given system of equations is $x, y = 3, 99$. Thus, the value of y is 99.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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